

Instructor Notes

Web Basics – HTML5

Lesson 00:

IGATE is now a part of Capgemini

People matter, results count.

Capgemini Public



Copyright © 2011 IGATE Corporation (a part of Capgemini Group). All rights reserved.

No part of this publication shall be reproduced in any way, including but not limited to photocopy, photographic, magnetic, or other record, without the prior written permission of IGATE Corporation (a part of Capgemini Group).

IGATE Corporation (a part of Capgemini Group) considers information included in this document to be confidential and proprietary.

Instructor Notes:**Document History**

Date	Course Version No.	Software Version No.	Developer / SME	Change Record Remarks
1-Oct-09	1.0	1.0	Kumar Bramhadande	Initial Document
May 2011	2.0		Karthik M	Integration Refinements
20-May-2013	3.0	4.0	Vinay Gupta	Revamped according to new curriculum
31-Mar-2015	4.0	5.0	Rathnajothi P	HTML material upgraded from version 4 to 5
May - 2016	4.1	5.0	Anjulata	Revamping as per new ToC
July – 2020	4.2	5.0	Neelima	Revamping as per new ToC

Capgemini Public



Keep this as a hidden slide.

Note to coordinators: Not to be printed for the class book.

Instructor Notes:

Course Goals and Non Goals

➤ **Course Goals**

- Understand Static web page creation.
- Create Static web pages.



➤ **Course Non Goals**

- DHTML (not covered in this course).

Instructor Notes:

Pre-requisites

➤ None

Instructor Notes:

Intended Audience

- Novice User
- Developer



Capgemini Public



Instructor Notes:

Day Wise Schedule

➤ **Day 1**

- Lesson 1: HTML Basics
- Lesson 2: Tables
- Lesson 3: Lists
- Lesson 4: Working with Links
- Lesson 5: Image Handling
- Lesson 6: HTML Forms for User Input
- Lesson 7: New Form Elements

Capgemini Public



Instructor Notes:

Table of Contents

➤ **Lesson 1: HTML Basics**

- Understand the structure of an HTML page.
- New Semantic Elements in HTML 5
- Learn to apply physical/logical character effects.
- Learn to manage document spacing.

➤ **Lesson 2: Tables**

- Understand the structure of an HTML table.
- Learn to control table format like cell spanning, cell spacing, border

➤ **Lesson 3: List**

- Numbered List
- Bulleted List

Instructor Notes:

Table of Contents

➤ Lesson 4: Working with Links

- Understand the working of hyperlinks in web pages.
- Learn to create hyperlinks in web pages.
- Add hyperlinks to list items and table contents.

➤ Lesson 5: Image Handling

- Understand the role of images in web pages
- Learn to add images to web pages
- Learn to use images as hyperlinks

Instructor Notes:

Table of Contents

➤ **Lesson 6: HTML Form for User input**

- Understand the role of forms in web pages
- Understand various HTML elements used in forms.
 - Single line text field
 - Text area
 - Check box
 - Radio buttons
 - Password fields
 - Pull-down menus
 - File selector dialog box

➤ **Lesson 7: New Form Elements**

- Understand the new HTML form elements such as
 - date
 - number
 - range
 - email
 - search
 - datalist
- Understand audio, video, article tags

Capgemini Public



Instructor Notes:

References

➤ Books

- Head First HTML with CSS
- HTML 5 in action
- HTML 5 and CSS 3
- Html Pocket Reference



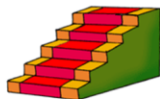
➤ Links

- <http://imbolc.ucc.ie/~pflynn/books/htmlcard.html>
- <http://www.w3.org/MarkUp/>

Instructor Notes:

Next Step Courses (if applicable)

- Working with any Scripting Language
- Working with Dynamic HTML



Capgemini Public



Instructor Notes:

Other Parallel Technology Areas

➤ None

Instructor Note

Web Basics - HTML

Lesson 1. HTML Basics

Capgemini Public



Lesson Objectives

➤ **After completing this lesson, you will be able to:**

- Understand the structure of an HTML page.
- Learn to apply physical/logical character effects.
- Learn to manage document spacing.
- New Semantic Elements in HTML 5



Capgemini Public



Instructor Note

1.1: Understand the structure of an HTML page

What is HTML?

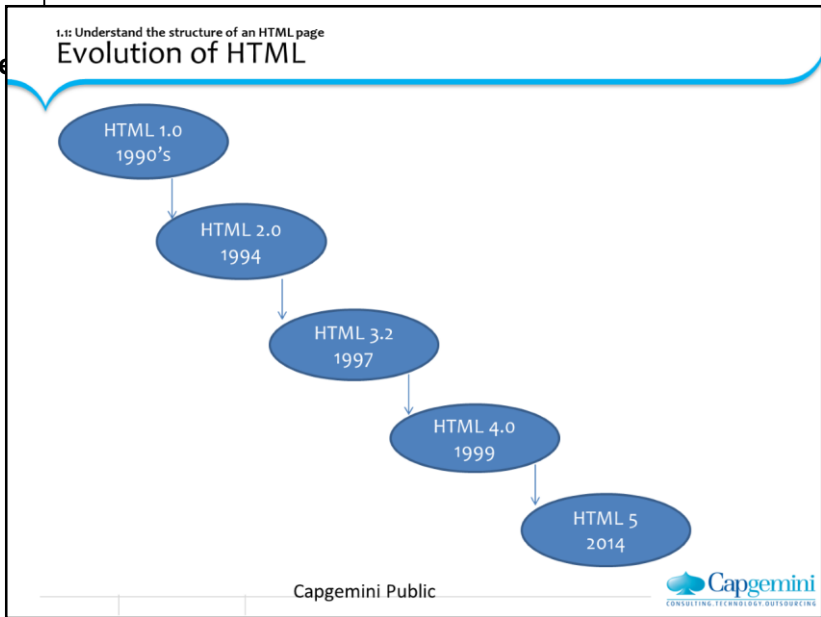
- **HTML is a language for describing web pages.**
 - It stands for Hyper Text Markup Language
 - HTML is a markup language and not a programming language
 - HTML uses markup tags to describe web pages.

Capgemini Public



HTML is a markup language and these documents describe web pages. Any HTML document contains HTML tags and plain text. HTML documents are also called web pages.

A web browser (like Internet Explorer or Firefox) reads HTML documents and display them as web pages. The browser does not display the HTML tags, but uses the tags to interpret the content of the page:



HTML is a markup language and these documents describe web pages. Any HTML document contains HTML tags and plain text. HTML documents are also called web pages.

A web browser (like Internet Explorer or Firefox) reads HTML documents and display them as web pages. The browser does not display the HTML tags, but uses the tags to interpret the content of the page:

Instructor Note

1.1: Understand the structure of an HTML page

HTML5 – The Good News !!!

- Good support on modern mobile devices (iOS, Android)
- Simpler, more intuitive syntax
- Video and Audio can be included without requiring a plug-in
- Incremental improvements to previous HTML challenges
- Much needed next step in HTML evolution
- Creative enhancements: Rounded corners, gradients, text layout
- Promising support of Mobile JS Frameworks (Sencha, jQTouch)

Capgemini Public



1.1: Understand the structure of an HTML page

HTML5 - Browser Support



Android 2.2



Chrome 6



Firefox 4.0



Safari Mobile



Opera 10.6



IE 9



Safari 5

Capgemini Public



Browser Support:

HTML5 is still a work in progress. However, most modern browsers have some HTML5 support. The latest versions of Safari, Chrome, Firefox, and Opera support some HTML5 features. Internet Explorer 9 will support some HTML5 features.

Some Statistics:

Chrome 10.0 is the big winner coming in. They have already implemented most of the functionality to be a compatible HTML5 browser. In addition, many of the features

that are not added are partially added. So Google Chrome is definitely way out in front in the race for HTML compatible browsers.

Firefox 4.0 is next in line. They are still missing a lot of key elements but got bonus points for the audio and video implementation as well as their parsing rules.

Internet Explorer 8 is pretty far behind the curve. Pretty dismal showing for what used to be the top internet browser in the world. Internet Explorer has been playing

catch up with rendering design since the implementation of CSS and their poor showing here tells us it does not seem like much will change in the future.

So those are the statistics. At the moment the only HTML5 browser that is going to

get you very far is Google Chrome and until browsers catch up with the newer language it is probably a good idea to use it sparingly in your designs until it actually

is a true and tested standard.

1.1: Understand the structure of an HTML page

HTML Elements

- Most Web documents are created using HTML.
- Documents are saved with extension .html or .htm.
- Tags are strings in the language surrounded by a less-than (<) and a greater-than (>) sign.
 - Opening tag: <html> Ending tag: </html>
- Can have Attributes
 - Attributes are Name-Value pairs added to HTML start tags.

Capgemini Public



HTML documents are text files made up of HTML elements. These files are saved with an extension of .htm or .html. A file with this extension indicates to the browser that the file is an HTML document.

You can check for the document structure and look for the presence of HTML element or tag. HTML elements are defined using HTML tags. Following are features of HTML tags:

Used to mark-up HTML elements.

Surrounded by the two characters "<" and ">".

Normally come in pairs.

HTML element starts with a start tag . Then, the content of the

HTML element is placed. This ends with an end tag

Not case sensitive.

For HTML, the usual filename extension is .html
(.htm for PC-based servers).

Q: Do we need to write the end tag for all the elements?

A: In HTML, there are some tags which do not have a closing tag. For example,
 and <hr> tags. So, it is not necessary to write the closing tags for such tags. But it is a good idea to write an empty tag by providing a "/" in the end tag.

1.1: Understand the structure of an HTML page

Rules applicable for HTML5 Tags

➤ Rules for HTML5 Tags are:

- The document must include with an HTML5 DOCTYPE.
- Tags and attributes are case-insensitive.
- Attributes do not need to be quoted.
- End tags are not required for every element.
- Some attributes may be empty such as checked and disabled.
 - For example, `<input type=checkbox checked>`
- Only void elements such as `br`, `img` and `link` may be "self-closed" with `/>`.

Capgemini Public



We stated that HTML tags are not case sensitive, but let's always use lowercase tags. To prepare yourself for the next generations of HTML, start using lowercase tags.

Tag Attributes

Tags can have attributes. Attributes provide additional information about HTML elements on your page.

For an Example:

This tag defines the `<body>` element of your HTML page. With an added `bgcolor` attribute, you can tell the browser that the background color of your page should be red, like this:

```
<body bgcolor="red">
```

Attributes always come in name/value pairs like this:

`name="value"`. Attributes are always added to the start tag of an HTML element.

Quote Styles

Always enclose values in quotes. Double quotes are more common, but single quotes are also allowed. In some rare situations, like when the attribute value itself contains quotes, it is necessary to use single quotes:

```
name='John "ShotGun" Nelson'
```


1.1: Understand the structure of an HTML page

HTML Elements (Code)

➤ Code Snippet

An HTML document appears as follows:

```

<!DOCTYPE HTML>
<html>
  <head>
    <title>Title of page</title>
  </head>
  <body>
    This is my first homepage. <b>This text is bold</b>
  </body>
</html>

```

Capgemini Public



Save the file as "mypage.html".

Click "Browse" (or "Choose File") and locate the HTML file you just created - "mypage.html". Select it and click "Open". Now you should see an address in the dialog box, for example "C:\MyDocuments\mypage.htm". Click OK, and the browser will display the page.

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0//EN">
```

It tells type, version, language of particular document. The first tag in your HTML document is <html>. This tag tells your browser that this is the start of an HTML document. The last tag in your document is </html>. This tag tells your browser that this is the end of the HTML document.

Text between the <head> tag and the </head> tag is the header information. This information is not displayed in the browser window. Text between the <title> tags is the title of your document. Title is displayed in your browser's title bar. Text between the <body> tags is the text that is displayed in your browser. Text between the and tags is displayed in a bold font.

1.1: Understand the structure of an HTML page

HTML Document Sections

- **HTML Head Section:**
 - `<head>...</head>`
 - Page Title, Base URL, Meta Information
- **HTML Body Section:**
 - `<body>...</body>`
 - Text, Images, Tables Colors, etc.

Capgemini Public



Head element can contain information about the document. The browser does not display this information to the user. Following tags can be in the head section: `<base>`, `<link>`, `<meta>`, `<script>`, `<style>`, and `<title>`.

Body element defines the document's body. It contains all the contents of the document (like text, images, colors, graphics, etc.).

Instructor Note

1.1: Understand the structure of an HTML page

Head Section: Page Title (Code)

➤ Code Snippet

Document Title is displayed using <title>...</title> tag.

```
<!DOCTYPE HTML>
<html>
  <head>
    < title>
      My First Page
    </ title>
  </head>
</html>
```

Capgemini Public



<TITLE> element does not have any visible effect within a browser's client area; however, the enclosed title appears in the title bar of the browser window.

1.1: Understand the structure of an HTML page

Demo

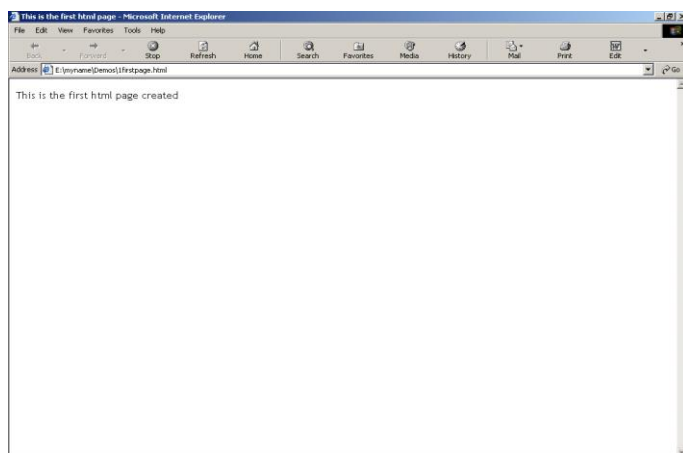
- Firstpage.html



Capgemini Public



```
<html>
<head><title>This is the first html page</title>
<body>This is the first html page created</body>
</head>
</html>
```



1.1: Understand the structure of an HTML page

Head Section: Document Base URL

➤ Code Snippet

```
<!DOCTYPE HTML>
<html>
  <head>
    <title> Document Base URL Manipulation </title>
    <base href="URL/">
  </head>
</html>
```

More Examples:

```
<base href="http://www.state.edu/images/">
<base href="ftp://ftp.state.edu/images/">
```

Capgemini Public



Base element specifies a base URL for all the links in a page.
 Note: The <base> tag must go inside the head element.

You can set the <base> element only once in a document, in the header. <base> element does not work outside the header.

Assume that the absolute address for an image is:

Now, we insert the <base> tag, which specifies a base URL for all of the links on a page, in its head section:

```
<head>
  <base href="http://www.state.edu/images/" />
</head>
```

When we insert images on the page, in the example above, we just specify the relative address, and the browser looks for that file using the full URL, "http://www.state.edu/images/smile.gif".

1.1: Understand the structure of an HTML page

Head Section: Meta Information

- The <meta> tag provides metadata about the HTML document such as descriptions and keywords for search engine.
- Metadata will not be displayed on the page, but will be machine parsable.
- Examples:

```
<meta .....>
<meta name="keywords" content="HTML, CSS" />
<meta name="author" content="Username">
<meta http-equiv=refresh content=60 />
<meta http-equiv=refresh content="20;url=c:/html/htm3.htm" />
<meta charset="UTF-8">
```

Capgemini Public



META: This element is used to specify additional meta information and appears in the <head> tag. You can specify multiple <meta> tags for the document. It provides meta-information about your page, such as descriptions and keywords for search engines and refresh rates. Following are some uses of meta:

Define keywords for search engines:

```
<meta name="keywords" content="HTML, DHTML, CSS, XML, XHTML, JavaScript, VBScript" />
```

Provide a description of your web page:

```
<meta name="description" content="Free Web tutorials on HTML, CSS, XML, and XHTML" />
```

Define the last revision of your page:

```
<meta name="revised" content="Hege Refsnes, 6/10/99" />
```

Refresh page every 5 seconds:

```
<meta http-equiv="refresh" content="5" />
```

Instructor Note

1.1: Understand the structure of an HTML page

Demo

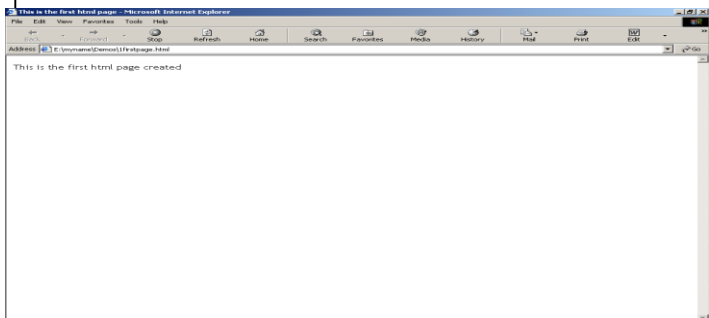
- Meta.html



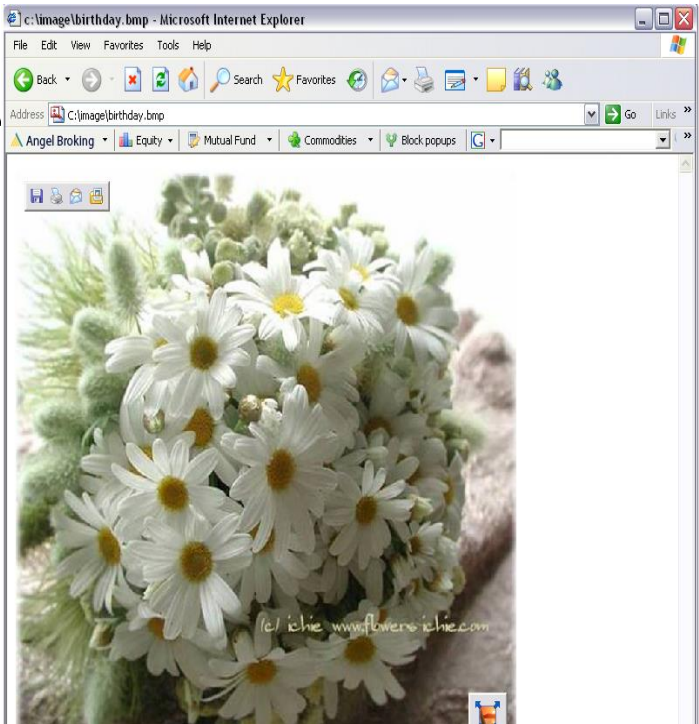
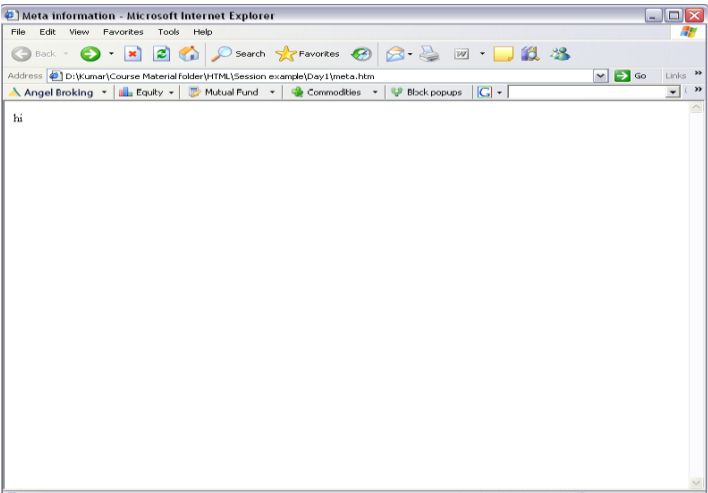
Capgemini Public



```
<!DOCTYPE HTML>
<html>
<head>
<!-- in meta tag in name attribute user can specify the value like keywords, description -->
<meta name="generator" content="microsoft visual studio 6.0">
<title>meta information</title>
<!-- this meta information help you to refresh the page after 2 sec and load image -->
<meta http-equiv=refresh content="2;url=c:\image\birthday.bmp"/>
</head>
<body>
Hi <p>&nbsp;</p>
</body>
</html>
```



Instructor Notes: Before Automatic Refresh



Instructor Note

BGCOLOR, BACKGROUND, TEXT, LINK, ALINK, VLINK are the attributes in body element which are obsolete in HTML5. Use CSS 3 for defining these attributes.

1.1: Understand the structure of an HTML page

HTML Body Section

➤ <body> Element:

- Represents information content.
- Each document can have at most one <body> element.
- Body element is placed between </head> and </html> elements.
- Attributes supported in <body> element are:
 - Event Handler attributes like ononline, onoffline, onunload, onpagehide, onpageshow, etc..
 - Global attributes like id, style, class, hidden, lang, etc..

Capgemini Public



<body> tag defines the HTML element containing the body of the HTML document.

1.1: Understand the structure of an HTML page

Document (Body) Contents

- **Body Text**
 - HTML truncates spaces in your text.
 - Use `<br>` to insert new lines.
 - Use `<p>` tag to create paragraphs.
- **Other Elements of Body Section:**
 - `<table>` tags are used to create tables.
 - `<img>` tags are used to insert images.

Capgemini Public



When you write HTML text, you are never sure how text is displayed in another browser. Some computers have large displays whereas some have small ones. Text is reformatted every time the user resizes the window.

Avoid reformatting text in your editor by adding empty lines and spaces. HTML truncates spaces. Any number of spaces as well as a new line counts as one space. Use `<br>` to insert blank lines. You might have noticed that `<br>` tags can be written without the closing tag `</br>`.

HTML automatically adds an extra blank line before and after some elements, like before and after paragraphs and headings.

Use a `<p>` tag to start a new paragraph, to which you can assign new attributes. Most browsers also place an extra space after a `<p>`. A `<br>` causes the browser to maintain the current paragraph attributes but to start placing text on a new line.

`</table>` is used to close the table tag.

To insert an image, the tag `` is used. Image should be saved with extension .gif. The `<img>` tag is empty, which means that it contains attributes only and it has no closing tag.

Instructor Note

1.1: Understand the structure of an HTML page

Comments in HTML Document

- Increase code readability.
- Ignored by the browser.
- Example of HTML comment:
 - <!-- This is a Sample HTML Comment -->

Capgemini Public



Comments are used for better code readability. Browser will ignore a comment. You can use comments to explain your code, or to store program-specific information, which can help you when you edit the source code at a later date.

Comments are not visible to the user, but they are still available to the program. A good practice is to comment text inside the script and style elements to prevent older browsers, that do not support scripting or styles, from showing it as plain text.

To include comments in the document, use the comment tag. Comment is a special tag starting with a "<" sign, followed by "!" and two hyphens. Then, type the commented text. It ends with two hyphens and a ">" sign.

1.1: Understand the structure of an HTML page

Demo

➤ Body.html



Capgemini Public



```
<html>
<head>
<title>body tag</title>
</head>
<body bgcolor="pink" text="red" alink="green" link="yellow">
<a href="body.html">background</a> color of the page is pink.
and text color is red
</body>
</html>
```

1.1: Understand the structure of an HTML page

Home Page Concepts

- Most popular home pages reflects the personality of the sponsoring organization or corporation.
- Keep the initial home page short and to the point.
- First element visitor sees is a collection of navigation buttons to navigate to other pages.
- When you publish a URL in print or any other marketing material, it points to the location of your home page.

Capgemini Public



1.2: physical/logical character effects

Physical Character Effects

- Tags applies physical character effects by formatting the characters are listed below:

Tag Name	Description	Example
<code>...</code>	To specify bold text	Language
<code><i>...</i></code>	Displays content in italic to define a part of text in an alternate voice or mood	<i>Language</i>
<code><u>...</u></code>	Tags make text between them underlined	<u>Language</u>
<code><s>...</s></code>	Tags make text between them appear striked out	Language
<code><sub>...</sub></code>	To specify subscript text	H ₂ O
<code><sup>...</sup></code>	To define superscript text	27 th May

Capgemini Public



BOLD

`<b>... </b>`: Tags make text between them bold e.g. `<b>capgemini</b>` is displayed as “**capgemini**”.

ITALIC

`<i>... </i>`: Tags make text between them italicized e.g. `<i>icapgemini</i>` is displayed as “*icapgemini*”.

UNDERLINE `<u>... </u>`: Tags make text between them underlined e.g. `<u>capgemini</u>` is displayed as “capgemini”.

STRIKETHROUGH

`<strike>` or `<s> .... </strike>` or `<s>`: Tags make text between them appear striked out e.g. `<s>capgemini</s>` is displayed as “~~capgemini~~” (with a strike across it.)

Subscripts and Superscripts

`<sub>text</sub>` subscripts text e.g. Chemical formula of water is H`<sub>2</sub>`O is displayed as “H₂O”:

Formula for a parabola is $y = x^2$ is displayed as “ $Y=X^2$ ”.

Instructor Note

1.2: physical/logical character effects

Demo

➤ PhysicalCharacterEffect.html



Capgemini Public



Instructor Note

1.2: physical/logical character effects

Logical Character Effects

- **Heading Styles:**
- `<hn>.....</hn>`
- Value of n can range from 1 to 6

```
<h1>This is level 1 heading</h1>
```

- **Syntax**

Capgemini Public



Heading Style

`<hn>.....</hn>`: Tag identifies headings and subheadings in a document. Here, n is the size of the heading ranging from 1 to 6 where 1 is largest and 6 is smallest.

For example:

```
<h1>This is an example of level 1 heading</h1>  
<h6>This is an example of level 6 heading</h6>
```

```
<html>  
<body>This is the first html page created  
<h1>This is level 1 heading</h1>  
<h2>This is level 2 heading</h2>  
<h3>This is level 3 heading</h3>  
</body>  
</html>
```


1.2: physical/logical character effects

Logical Character Effects (Contd...)

<code>	Displays any code part in the web page.
<var>	Displays any variable on the web page.
<kbd>	Displays computer commands and arguments.
	Highlights specific areas of text enclosed within it.
	Used for defining important text.
<dfn>	Designed specifically for words and phrases defined in text.
<address>	Specifies information about the creator e.g. Address, mailing address.

Capgemini Public



Code: `<code>...</code>`: Tag displays any code part on the web page.

e.g. `<code>perform 2000-modify-para</code>` displays
“perform 2000-modify-para”

Var: `<var>...</var>`: Tag displays any variable on the web page

e.g. `<var>count</var>` displays count.

Keyboard

`<kbd>...</kbd>`: Tag to display computer commands and arguments, especially those to be entered by the user. The text tagged by `<kbd>` is usually shown in a fixed-width font. Indicating text as keyboard input.

E.g. `<kbd>copy *.exe c:\dir1</kbd>` displays

copy *.exe c:\dir1.

Instructor Notes:

Emphasis

`<em>.....</em>`: Tag provides generic emphasis or highlights specific areas of text without requiring the browser to set a specific physical style for them e.g. `<em>This is capgemini</em>` displays “*This is capgemini*”.

Strong

`<strong>... </strong>`: Tag for strong emphasis e.g. `<strong>This is an L2 question</strong>` displays “This is an L2 questions”.

Emphasis and *strong emphasis* tags highlight specific areas of text without the browser having to use a specific physical style for them.

Definition

`<dfn>.....</dfn>`: Tag is designed specifically for words and phrases that are defined in the text. Physical method of emphasizing the words is left to the browser. This tag changes the style of the text contained between the `<dfn>` and `</dfn>` tags. Style used is dependent on the browser, but is usually either bold or italic e.g.

`<dfn>Definition</dfn>`.

Address

`<address>... </address>`: Tag is used to specify information about the creator or maintainer of a Web page. This information can include the name, e-mail address, phone number, mailing address, or other relevant information.

Special Characters in HTML

➤ Character Entities

- Comprise following three parts:
 - Ampersand (&),
 - Entity name or a #
 - Character code
 - Semicolon (;)
- Included in HTML page using:
 - Character code/Entity number: Include any character using its ISO Latin 1 character code.
 - To display ">" symbol, character code is 62 i.e. >

Capgemini Public



Some characters like the "<" character, have a special meaning in HTML, and therefore cannot be used in text. To display a less than sign (<) in HTML, we need use a character entity.

Character Entities

Some characters have a special meaning in HTML, like the less than sign (<) that defines the start of an HTML tag. If we want the browser to actually display these characters we must insert character entities in the HTML source.

A character entity has three parts: an ampersand(&), an entity name or a # and an entity number, and finally a semicolon (;). To display a greater than sign in an HTML document we write: > or >

The advantage of using a name instead of a number is that a name is easier to remember. The disadvantage is that not all browsers support the newest entity names, while the support for entity numbers is very good.

Note: Entities are case sensitive.

Non-Breaking Space

The most common character entity in HTML is the non-breaking space. Normally HTML will truncate spaces in your text. If you write 10 spaces, HTML will remove 9 of them. To add spaces, use the character entity.

Instructor Note

1.3: Managing Document Spacing

Managing Document Spacing

- **Horizontal Spacing**
 - Use `<hr>` tag for including horizontal rule in an HTML document
- **Vertical Spacing**
 - `<p>`: Paragraph Break
 - `<br>`: Line Break

Capgemini Public



`<p> ... </p>`: Tag to start and end the new paragraph.

`<br>... </br>` Tag maintains current paragraph but enters text in the new line.

Instructor Note

1.3: Managing Document Spacing

Demo

- LCharacterEffects.htm
- Spacing.html



Capgemini Public



```
<html>
<head><title>Horizontal and Vertical spacing</title></head>
<body>
<p>
This paragraph contains a lot of lines
in the source code,
but the browser ignores it.
</p>
<hr size="2" width="50%" color="blue">
<p>
Notice the horizontal rule occupying 50 % of the window width.
</p>
This paragraph contains <br> line breaks in the
source code <br> so this
is the third line displayed within the paragraph.
</body>
</html>
```

Instructor Note

1.3: Managing Document Spacing

Preformatted Text (Code)

- `<pre>...</pre>` Tags display preformatted blocks of text with a fixed-space font.
- `<pre>` tag displays text with white space, line breaks, and tabs.
- Examples:

```
<!DOCTYPE html>
<html>
  <head><title> Preformatted Text</title>
  <body>
    <pre>Browser would display this paragraph
    as you are viewing here.
    No need to provide line or paragraph breaks. </pre>
  </body>
</html>
```

Capgemini Public



`<pre>...</pre>` Tags display preformatted blocks of text with a fixed-space font. When they appear inside `<PRE>` tag, white space, line breaks, and tabs are also displayed. It is mainly useful for simulating program listings.

Note: The `<pre>` tag takes care of carriage returns, tabs, spaces and even the bold tag.

E.g.:

```
<html>
<head><title>PREFORMATTED TEXT EXAMPLE</title></head>
<body>
<h3>EMPLOYEE DETAILS PREFORMATTING</h3>
<hr>
<pre>
<b>EMPLOYEE DETAILS</b>
EMPID      EMPNAME      DEPT
1001      JOHN      ACCOUNTS<BR>
1002      TOM      PURCHASE<BR>
1003      TAMMY      SALES<BR>
</pre>
</body></html>
```

Instructor Note

1.3: Managing Document Spacing

Divisions in an HTML Document (Code)

- The <div> tag defines a division or a section in an HTML document.
- The <div> tag is used to group block-elements to format them with CSS.
- Syntax:

```
<div>.....</div>
```

- Example:

```
<!DOCTYPE html>
<html>
<body>
<div style="text-align: center">Text is center aligned.</div>
<div style="text-align: left">Text is left aligned.</div>
</body>
</html>
```

Capgemini Public



<div>... </div>: Tag creates divisions in Web pages. These can be used to set the alignment for an entire section of the page.

<div align=center>This text is at the center of the browser window.</div>:

Tag to align the text to the center of web page.

<div align=left>This text is at the left side of the browser window.</div>:

Tag to align the text to the left of the web page.

<div align=right>This text is at the right side of the browser window.</div>: Tag to align the text to the right of the web page.

Difference Between **pre** and **div** Tags:

<pre> is used for formatting the entire text.

<div> is used to divide the webpage and use center, left or right attribute to align the text of the divided web page.

Difference Between **div** and **span** Tags:

div is used as a paragraph break as it creates a logical division of the document. In contrast, SPAN simply applies style and alignment as specified.

div has **align** attribute in it which is not present in case of **span**.

div is generally used for a block of text. **span** is used for words or sentences.

Primary difference between them is that does not format by itself. The <div> tag includes a paragraph break, as it defines a logical division in the document.

Instructor Note

1.3: Managing Document Spacing

 in an HTML Document

- The HTML tag is used for grouping and applying styles to inline elements.
- The span tag is used with inline elements whilst the div tag is used with block-level content.
- Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

HyperText Markup Language is the standard markup language used to create

```
<span style="color:blue;font-weight:bold"> static </span> web pages </body>
```

```
</html>
```

Capgemini Public



SPAN is used for words or sentences.

Primary difference between them is that does not format by itself. The <div> tag includes a paragraph break, as it defines a logical division in the document.

Instructor Note

Block level element

- Block elements separate content into blocks
- Each block element is displayed on its own with a line break before and after it
- Example for block elements are `<p>`, `<div>`, `<ol>`, `<ul>`, `<li>` etc..
 - `<div>` element is a block level element that can be used as a container for other HTML elements

Capgemini Public



Inline element

- Inline elements won't separate content into blocks
- Inline elements are normally displayed without line breaks.
- Inline element is just displayed in the flow of the paragraph
- Example for Inline elements are `<b>`, `<i>`, `<u>`, `<em>`, `<a>`, `<td>`, `<img>`, etc..
 - `<span>` element is an inline element that can be used as a container for text.

Capgemini Public



Instructor Note

Rules

- All inline elements and text need to be nested inside another block element before they can go in the `<body>` element
- Block elements are not allowed inside an inline element
- Keep block elements out of your `<p>` element
- Put text and inline elements inside block elements before adding them to a `<blockquote>`
- For an example:
 - `<ol>` and `<ul>` can have only `<li>`
 - We can put text, inline elements or block elements inside `<li>`

Capgemini Public

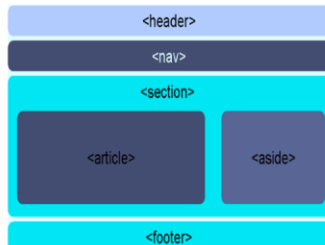
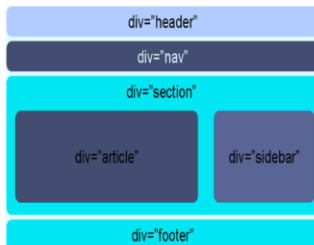


Example of Inline and Block elements

```
<!DOCTYPE html>
<html>
<body>
<p>HyperText Markup Language is the standard markup language
used to create <span style="color:blue;font-weight:bold"> static
</span> web pages. JavaScript is a scripting language used to make
web page content as <span style="color:orange;font-
weight:bold">dynamic</span>.</p>
</body>
</html>
```

Laying out a page with HTML5

- Most HTML 4 pages include a variety of common structures, such as headers, footers and columns
- It's common to mark them up using div elements, giving each a descriptive id or class
- HTML 5 addresses this issue by introducing new elements for representing each of these different sections
- Elements that make it much easier to structure pages



Capgemini Public

HTML5 Page Structure

HTML 5 introduces a whole set of new elements that make it much easier to structure pages. Most HTML 4 pages include a variety of common structures, such as headers, footers and columns and today, it is fairly common to mark them up using `div` elements, giving each a descriptive id or class. The use of `div` elements is largely because current versions of HTML 4 lack the necessary semantics for describing these parts more specifically. HTML 5 addresses this issue by introducing new elements for representing each of these different sections. The `div` elements can be replaced with the new elements: `header`, `nav`, `section`, `article`, `aside`, and `footer`.

Page Simplification

1. Small degree of freedom, but very detailed rule for parsing
2. Validation will matter
3. HTML5 is designed so that old HTML4 browsers can safely ignores new HTML5 constructs
4. Documents must not use deprecated features –extensive error handling by user agents

New Semantic Elements in HTML 5

| HTML Tag | Description |
|-----------|---|
| <article> | Defines a complete, self-contained block of related elements. |
| <aside> | Defines some content aside from the content it is placed in (like a sidebar) |
| <footer> | Defines a footer for a document or section which contains the author of the document, copyright information, links to terms of use, contact information |
| <header> | Defines a header for a document or section. It can be used as a container for introductory content. |
| <nav> | Defines a set of navigation links |
| <section> | Defines a section in a document |

Demo

➤ LayoutDemo.html



Capgemini Public



```
<html>
<head><title>Horizontal and Vertical spacing</title></head>
<body>
<p>
This paragraph contains a lot of lines
in the source code,
but the browser ignores it.
</p>
<hr size="2" width="50%" color="blue">
<p>
Notice the horizontal rule occupying 50 % of the window width.
</p>
This paragraph contains <br> line breaks in the
source code <br> so this
is the third line displayed within the paragraph.
</body>
</html>
```

Instructor Note

Lab Session

> Lab 1



Capgemini Public



Summary

- **In this lesson, you have learnt about**
- The structure of an HTML page.
 - Physical/logical character effects
 - Managing document spacing.
 - New Semantic elements in HTML5



Capgemini Public



Instructor Note

Answers

1. Option 1 and 3
2. False
3. variable

Review Question

➤ **Question 1: Which of the following are newly added content tags in HTML5?**

- Option1: article
- Option2: div
- Option3: section
- Option4: p



➤ **Question 2: HTML document is saved with an extension .xml.**

- True/False

➤ **Question 3: A Var tag is used to display the _____ in the web page.**

Capgemini Public



Instructor Note

Review Question: Match the Following

Answers

- 1 - c
- 2 - d
- 3 - a
- 4 - f
- 5 - b
- 6 - e

| | |
|---------------|---|
| 1. Code | a) Tag displays user-entered computer commands and arguments |
| 2. Var | b) Tag is used for defining important text. |
| 3. Kbd | c) Tag displays any code part on the web page. |
| 4. Emphasis | d) Tag displays any variable on the web page. |
| 5. Strong | e) This tag designed specifically for words and phrases that are defined in the text. |
| 6. Definition | f) Tag for generic emphasis means this tag highlight specific areas of text. |



Capgemini Public

Instructor Notes

Web Basics – HTML5

Lesson 2. Tables

Capgemini Public



Instructor Notes

Lesson Objectives

- **After completing this module, you will be able to:**
 - Understand the structure of an HTML table.
 - Controlling table format



Capgemini Public



Instructor Note:

2.1: Tables

Creating Tables

- Table contains data in the format of rows and columns.
- For an example, department information's are displayed in the tabular format as shown below

| Deptno | Dname | Location |
|--------|------------|----------|
| 10 | Accounting | New York |
| 20 | Research | Dallas |
| 30 | Sales | Chicago |
| 40 | Operations | Boston |

→ Row

↓
Column named as "Deptno"

- The above "Department" table contains 4 rows and 3 columns.

Capgemini Public



Tables

Tables are defined with the <table> tag. A table is divided into rows (using <tr>), and each row is divided into data cells (using <td>). Letters td stand for "table data," which is the content of a data cell. A data cell can contain text, images, lists, paragraphs, forms, horizontal rules, tables, etc.

Instructor Note:

2.1: Tables

Creating Tables

An HTML table can be created using <table> elements

- <table>
 - Define an HTML table
 - Other elements like <tr>, <caption>,... can be nested inside <table> element

An HTML table has two kinds of cells

- Header Cells
 - <th>
 - Defines a table header
- Standard Cells
 - <tr>
 - Defines a table row
 - A row can have one or more <td> or <th> elements
 - <td>
 - Defines a table cell data

| Employee Name | Salary |
|---------------|--------|
| Lathika | 55000 |
| Sandeep | 32000 |
| Ajay | 99322 |

Header Cells

Standard Cells

Capgemini Public



An HTML table has two kinds of cells:

- Header Cells: Contain header information (created with the `th` element).
- Standard Cells: Contain data (created with the `td` element).
- The text in a `th` element is bold and centered.
- The text in a `td` element is regular and left-aligned.

<tr> stands for Table row

<td> stands for table data

<th> stands for table header

Instructor Notes

2.1: Tables

Creating Tables

> Syntax

```
<table>
  <tr> <th>Column1 Header</th> <th>Column2 Header</th></tr>
  <tr> <td>Cell 1,1</td> <td>Cell 1,2</td> </tr>
  <tr> <td>Cell 2,1</td> <td>Cell 2,2</td> </tr>
</table>
```

Capgemini Public



Creating Tables

➤ **Some more elements which can be used while creating tables are:**

- **<thead>**
 - Group header content in an HTML table
- **<tbody>**
 - Group the body content in an HTML table
- **<tfoot>**
 - Group footer content in an HTML table
- **<caption>**
 - Defines a caption for the table
 - <caption> element should follow with <table> element immediately.
 - <caption> element value will be center aligned and displayed above the table

The table also has a few more tags to layout your data.

The `<thead>` tag is used to group the header content in an HTML table. The `<thead>` element should be used in conjunction with the `<tbody>` and `<tfoot>` elements.

The `<tbody>` element is used to group the body content in an HTML table and the `<tfoot>` element is used to group the footer content in an HTML table.

`<tfoot>` must appear before `<tbody>` within a table, so that a browser can render the foot before receiving all the rows of data.

Note that the `<thead>`, `<tbody>` and `<tfoot>` elements are seldom used, because of bad browser support. Expect this to change in future versions.

Instructor Note:

2.1: Tables

Creating Tables

➤ Syntax

Table column headings:

```
<table>
  <caption>This is table caption</caption>
  <tr>
    <th>COLUMN 1</th>
    <th>COLUMN 2</th>
    <th>COLUMN 3</th>
  </tr>
</table>
```

Capgemini Public



Table Caption:

- The <caption> tag defines a table caption.
- The <caption> tag must be inserted immediately after the <table> tag.
- You can specify only one caption per table. Usually the caption will be centered above the table.

Table Headers:

The <th> tag defines a header cell in an HTML table.

Instructor Notes

2.1: Tables

Creating Tables

Syntax

```
<table>
<thead>
<tr><td>.....</td></tr>
</thead>
<tfoot>
<tr><td>.....</td></tr>
</tfoot>
<tbody>
<tr><td>.....</td></tr>
<tbody>
</table>
```

Capgemini Public



Instructor Note

2.11: Tables

Blank Data Cell (Code)

➤ Inserting Blank Data Cell :

- `<td></td>`
- `<td><br></td>`

Capgemini Public



Blank Data Cell:

You can avoid missing a border around empty cells. Add a non-breaking space () to empty data cells, to make the borders visible.

Instructor Notes

2.1: Tables
Creating Tables

➤ Tableheading.html



Instructor Notes

2.1: Tables

Nested Tables (Code)

- A cell can contain another table within it.

```

<table>
<tr> <th>Zone</th> <th> State </th> </tr>
<tr> <td>South</td>
<td><table>
<tr> <th>Name</th> <th>Capital City</th> </tr>
<tr> <td>Karnataka</td> <td>Bangalore</td> </tr>
<tr> <td>Tamilnadu</td> <td>Chennai</td> </tr>
<tr> <td>Andhra Pradesh</td> <td>Hyderabad</td>
</tr>
</table></td></tr>
</table>

```

| Zone | State | |
|-------|----------------|--------------|
| | Name | Capital City |
| South | Karnataka | Bangalore |
| | Tamilnadu | Chennai |
| | Andhra Pradesh | Hyderabad |

Capgemini Public



Nested Tables:

Nesting tables within tables lets you create some complex effects from just basic HTML table tags. Nesting tables places one table inside of another. To nest a table, start the table as you normally would. When you come to the cell into which you want to place the nested table, enter the table data tag. Then, instead of typing the contents of the cell as you normally would, start the nested table. The table becomes that cell's content. After you end the nested table, be sure to end the table cell it is in with the </td> tag.

Instructor Notes

2.1: Tables

Demo

- tabnest4.htm



Capgemini Public



```
<html>
<head><title>Nested Tables</title></head>
<table width=100% height=100%>
<tr><th bgcolor="red">col1<th bgcolor="green">col2
<tr align=center><td bgcolor="aqua">data1
<td bgcolor="black"><font color="yellow">data2</font>
<table width=100% height=100%>
<tr><th bgcolor="pink">col1<th bgcolor="blue">col2
<tr align=center><td bgcolor="orange">data1<td bgcolor="purple">
<font color="yellow">data2</font>
</table>
</table>
</html>
```

Instructor Note:

2.2: Table Formatting

Table Formatting

➤ Cell Spanning

- Table cells can span across more than one column or row.
- Types of cell spanning
 - Row spanning
 - Column spanning

➤ Example of Colspan

| Employee Name | Salary |
|---------------|--------|
| HR | |
| Lathika | 55000 |
| Sandeep | 32000 |
| SALES | |
| Sonia | 43000 |
| Raju | 30022 |
| Ajay | 99932 |

➤ Example of Rowspan

| Department Name | Employee Name | Salary |
|-----------------|---------------|--------|
| HR | Lathika | 55000 |
| | Sandeep | 32000 |
| Sales | Sonia | 43000 |
| | Raju | 30022 |
| | Ajay | 99932 |

Capgemini Public



Cell Spanning:

There are two types of cell spanning. Row and column spanning.

Cell Spanning (Code)

➤ Row spanning/Column spanning:

- Use rowspan and colspan attribute either in <td> or <th> element.

```
<table>
  <tr>
    <th rowspan=m>Multiple Column Header</th>
    <th colspan=n>Multiple Row Header</th>
  </tr>
</table>
```

➤ m & n are integers specifying number of rows and columns respectively.

colspan="number of columns" ~

By default, the number of columns in a table is defined by the number of table data cells appearing in the table row that contains the most data. You would, typically place the same number of data cells in each table row. If a table row does not contain the requisite number of table cells, then it will essentially be in 'error' and will be displayed with a missing cell.

rowspan="number of rows" ~

Rowspan attribute works just like the colspan attribute except that you may find the situation a little more difficult to visualize when working with the source code. But once again the principle is the same. By using the rowspan attribute, you can force a table cell to span the number of rows specified by the respective value.

Cell Spanning Demo

➤ Table-span.html

| Deptno | Name | Salary |
|--------------|---------|--------|
| 10 | Sita | 34000 |
| | Ram | 24000 |
| 20 | Laksman | 45000 |
| | Lava | 32000 |
| | Kusha | 42000 |
| Total Salary | | 1222 |



Instructor Notes

2.2: Table Formatting

Grouping of Columns

- `<colgroup>` tag specifies a group of one or more columns in a table for formatting
- The `<col>` tag specifies column properties for each column within a `<colgroup>` element.
- Use `<colgroup>` and `<col>` tags to group columns with common properties like
 - Span attribute :
 - Identifies number of columns in the current group.
 - Default value is 1
 - Provide span attribute and omit `<col>` tag
- Example for grouping 3 columns and applying background color as green

```
<table>
  <colgroup span="3" style="background-color:green"> </colgroup>
  <col>
  <col>
  <tr>
    table contents.....
  </tr>
</table>
```

Capgemini Public



`<colgroup>` and `<col>` are another attempt to help define parts of a table more easily. Before these tags were defined, you had to keep track of how many `<td>` cells were in each row to know how many columns were in your table, which cells were in the same column, and how to format the cells in a given column.

Span: Defines the number of columns the `<colgroup>` should span. If you wish to put the first three columns in a group and format them the same way you need to write the following HTML code:

```
<table>
  <colgroup span="3"> </colgroup>
  <col>
  <col>
  <tr>
    table contents.....
  </tr>
</table>
```

`<col>` tag defines attribute values for one or more columns in a table. You can only use this element inside a `colgroup`. Use this element when you specify different attribute values to a column inside a `colgroup`. Without a `col` element a column will inherit all its attribute values from the column group.

Instructor Notes

2.2: Table Formatting

Demo

- tabcol3.htm



Capgemini Public



```
<html><head><title>Table with Colgroup</title></head>
<body>
<table width="100%" frame="void">
<tr><th colspan="3">Ecommerce Stream</th>
<th colspan="4">Mainframe Stream</th>
<colgroup span="3" width="20%" bgcolor="orange">
<colgroup span="4" width="10%" bgcolor="lightgreen">
<tr><td>Internet/HTML<td>Javascript<td>Java
<td>CICS<td>COBOL<td>JCL<td>IMS
<tr><td>JSP<td>Servlet<td>Struts
<td>ADS/O<td>VSAM<td>IDMS<td>MVS
</table>
</html>
```

Instructor Notes

Lab

- Lab 2



Capgemini Public



Instructor Note

Lesson Summary

- After completing this module you know:
 - Structure of an HTML table
 - **Control** table format such as cell spanning
 - Use tables to format contents of an HTML Page.



Capgemini Public



Instructor Notes

Answers

Question 1:

True

Question 2:

True

Question 3:

<tr> and <td>

Review - Questions

- Question 1: Cell spanning is used to joining cells together to make a larger cell.
 - True/False
- Question 2: The <col> tag defines the attribute values for one or more columns in a table.
 - True/ False
- Question 3: In which tag usage of rowspan/colspan attribute is valid?
 - <th>
 - <tr>
 - <td>
 - None of the above



Capgemini Public

 **Capgemini**
CONSULTING TECHNOLOGY OUTSOURCING

Instructor Note

Web Basics – HTML 5

Lesson 3. List

Capgemini Public



Instructor Note

Lesson Objectives

- In this lesson, you will learn about:
 - Numbered List
 - Bulleted List



Capgemini Public



HTML provides support for ordered, unordered and other types of lists

Instructor Note

3.1: Numbered List

Numbered List

- List of items can be numbered or ordered is called numbered/ordered List
- Tags can be used for creating ordered list are:
 -
 -
- tag
 - Defines a list item.
 - Should be used as a child tag of either or tag.
- tag
 - defines an ordered list.
 - An ordered list can be numerical or alphabetical.
 - Attributes in tag are
 - start
 - type

Capgemini Public



Numbered List:

HTML enables you to create a numbered, or ordered list that will automatically generate numbers in front of each item in the list.

The number placed in front of an item depends on the location of the item in the list.

At the beginning of the list, place the (for ordered list) tag.
 At the end of the list, place the tag and in front of each item in the list, place the (for list item) tag.
 End tag for is optional.

Instructor Note

3.1: Numbered List

Example

➤ Example (Code)

```
<!DOCTYPE html>
<html>
  <body>
    <h2>My favorite cricket teams</h2>
    <ol>
      <li>INDIA</li>
      <li>SRILANKA</li>
    </ol>
  </body>
</html>
```

➤ Output

My favorite cricket teams

1. INDIA
2. SRILANKA

Capgemini Public



3.1: Numbered List

Attributes in tag

- **Type Attribute:**
 - Specifies the kind of marked to use in the list
 - Valid values for the Type Attribute
 - A-Uppercase letters
 - a-Lowercase letters
 - I-Uppercase Roman letters
 - i-Lowercase Roman letters
 - 1-Standard numbers, default
- **Start Attribute**
 - Specifies the start value of an ordered list

Capgemini Public



You can even select the type of numbering system to be used with the type attribute. For example:

- A-Uppercase letters: <ol type=A>
- a-Lowercase letters: <ol type=a>
- I-Uppercase Roman letters: <ol type=I>
- i-Lowercase Roman letters: <ol type=i>
- 1-Standard numbers: <ol type=1>

Type attribute enables you to alter the style of numbers used in the lists, whereas start attribute sets the starting number of the list.

To start an ordered list at a number other than 1, place the **start** attribute on the tag at the beginning of the list.

You can also use the VALUE attribute within tag to change the numbering sequence within a list.

Instructor Note

3.1: Numbered List

Example

Example:

➤ Output

```
<!DOCTYPE html>
<html>
  <body>
    <h2>My favorite cricket teams</h2>
    <ol start="2" type="a">
      <li>INDIA
      <li>SRILANKA
      <li>PAKISTAN
      <li>AUSTRALIA
      <li>SOUTH AFRICA
    </ol>
  </body>
</html>
```

My favorite cricket teams

- b. INDIA
- c. SRILANKA
- d. PAKISTAN
- e. AUSTRALIA
- f. SOUTH AFRICA

Capgemini Public

 **Capgemini**
ÉNERGIE TECHNOLOGIE SERVICES

Instructor Note

3.1: Numbered List

Demo

➤ `NumberedList.html`



Capgemini Public



Bulleted List

- List of items prefixed with bullets is called Bulleted /Unordered List
- ** tag**
 - defines an unordered list.
 - An unordered list items will be marked with bullets (small black circles).
- **Example**

```
<!DOCTYPE html>
<html>
  <body>
    <h2>My favorite cricket teams</h2>
    <ul>
      <li>INDIA</li>
      <li>SRILANKA</li>
    </ul>
  </body>
</html>
```

➤ **Output**

My favorite cricket teams

- INDIA
- SRILANKA

Capgemini Public



To create a bulleted list use the (unordered list) tag at the opening of the list and at the end of the list.

3.2: Bulleted List

Attributes in tag

- **Style Attribute:**
 - Can be used in tag to define the style of the marker.

| Style Attribute Value | Description |
|------------------------------------|--|
| list-style-type:disc | The list items will be marked with bullets (default) |
| list-style-type:circle | The list items will be marked with circles |
| list-style-type:square | The list items will be marked with squares |
| list-style-type:none | The list items will not be marked |
| list-style-image: url('logo.gif'); | The list items will be marked with an image |

- **Example**
 - <UL style="list-style-type:circle"> will be used to prefix unordered list of items with square.

3.2: Bulleted List

Example (Code):

➤ Example

```
<!DOCTYPE html>
<html>
  <body>
    <h2>My favorite cricket teams</h2>
    <ul style="list-style-type:square">
      <li>INDIA</li>
      <li>SRILANKA</li>
    </ul>
  </body>
</html>
```

➤ Output

My favorite cricket teams

- INDIA
- SRILANKA

Capgemini Public



Instructor Note

3.2 Bulleted List
Demo

➤ **BulletedList.html**



Capgemini Public



Instructor Note

Lab

- Lab 3



Capgemini Public



Instructor Notes

Lesson Summary

- In this lesson, you have learnt about:
 - Types of available list items
 - Create bulleted, numbered and glossary listed items



Capgemini Public



Instructor Notes

Answers:

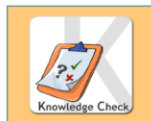
Question1:

Question 2:
True

Question 3:

Review Question: Questions

- Question 1: Which tag is used at the beginning of an Ordered List
 -
 -
 -
- Question 2: To create a bulleted list tag is used
 - True/ False
- Question 3: _____ attribute is used to modify the appearance of bullets



Capgemini Public



Instructor Note

Answers:

- 1 – e
- 2 – f
- 3 – d
- 4 – c
- 5 – a
- 6 – b

Review Question: Match the Following

| | |
|--|------------------------------|
| 1. <dd> | a. 1-Standard numbers |
| 2. <dt> | b. Square Form |
| 3. <ol type=l> | c. A-Uppercase letters |
| 4. <ol type=A> | d. I-Uppercase Roman letters |
| 5. <ol type=1> | e. Data Definition |
| 6. <ul style="list-style-type:circle"> | f. Data Term |



Capgemini Public


 The Capgemini logo, consisting of a blue cloud-like shape followed by the word 'Capgemini' in blue, with the tagline 'CONSULTING TECHNOLOGY AUTOMATION' in smaller blue letters below it.

Instructor Notes

Web Basics – HTML5

Lesson 4. Working with Links

Capgemini Public



Lesson Objectives

Instructor Notes

- **After completing this module, you will be able to:**
 - Understand the working of hyperlinks in web pages.
 - Learn to create hyperlinks in web pages.
 - Add hyperlinks to list items and table contents.



Capgemini Public



Instructor Notes

4.1 Working of hyperlinks in web pages

Hyperlink in HTML

- Links allow users to navigate and find information on the Internet as easy as possible.
- HTML links are hyperlinks.
- A hyperlink is a text or an image you can click on, and jump to another document.
- Tag to create a link is `<a>` which stands for anchor.
- Syntax:

```
<a href="url">link text</a>
```

- By default, links will appear as this in all browsers:
 - An unvisited link is underlined and text is highlighted in blue color ([Click me](#))
 - A visited link is underlined and text is highlighted in purple color ([Click me](#))
 - An active link is underlined and text is highlighted in red color ([Click me](#))

Capgemini Public



Hyperlinks make navigating and finding information on the Internet as easy as possible. When information is published on the Internet, hyperlinks allow readers to access related information on other Web pages or Web sites.

What makes HTML Hyper?

Hyper means more than or outside of. A hyperlink takes you to another page that gives you more than what was on the original page. That page is outside the main page, although it is connected to it.

Create Link to a Local Page**Anchor Tag**

Tag to create a link is `<a>`, which stands for anchor. Anchor tag begins with `<s>` and is followed by a reference comment (href) and the name of the page or the URL where the page to link resides. After closing the initial anchor command, you type in the text you wish to highlight as a link, and then close the tag with `</a>`.

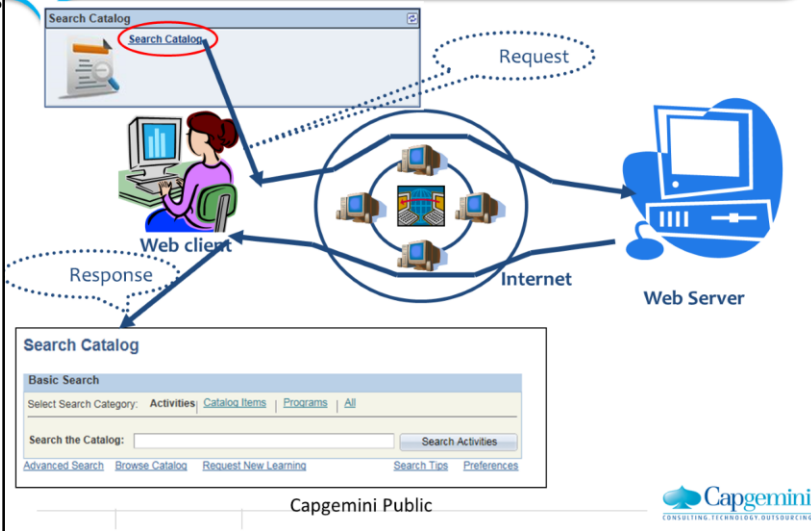
E.g.: The `<a href="news.htm">news </a>` on Indian Festival.

The link above displays the word news in blue with an underline. When you click it, you see the web page named news.htm.

Instructor Notes

4.1 Working of hyperlinks in web pages

How does Hyperlink works?



How does hyperlink works?

In the given example, when user clicks on search catalog link request is sent to the server via internet. In the server, request is processed and “Search catalog” page as response sent back to the web client for users view.

Instructor Notes

Discuss about Absolute and Relative linking. Absolute Linking: Entire address of the page.

Relative Linking: Only filename has to be specified.

4.2 create hyperlinks in web pages

Create Links

➤ Link to local document in an application(Relative linking)

```
<a href="contactus.html">Click Here</a>
```

➤ Link to local document in an application(Absolute linking)

```
<a href="http://localhost:8080/Example/contactus.html">Click Here</a>
```

➤ Link to web document

```
I have some <a href="http://www.state.edu/info/info.htm"> information</a> about education.
```

Capgemini Public



Create Link to Other Pages

Use Hypertext Transfer Protocol (HTTP), to connect to one computer and transfer a copy of documents from that computer to a browser that runs on another computer. Place the document URL in the anchor of a link using the **href** attribute.

For example:

I have some <a href=<http://www.stateu.edu/info/info.txt>> information about education

Instructor Notes

4.2 create hyperlinks in web pages

Links to an E-Mail

- Link to an electronic mail:
 - An URL tag called “mailto” identifies the address to which an e-mail has to be sent.
 - Syntax:

```
<a href="mailto:mailid?subject=content&cc=cc-mailid&bcc=bcc-mailid">Link
text</a>
```

- Examples:

```
<a href="mailto:eDude@igate.com?subject=More Info">Contact Me</a>
Please <a href="mailto:author@igate.com">mail</a> your comments to me
```

Capgemini Public



Create Link to an Electronic Mail

HTML includes a URL tag called mailto, which identifies the address to which an e-mail is to be sent to. If the link is selected, the browser starts an email program to send a message to the recipient listed in the URL.

For example:

Send me e-mail with your comments about my page.

4.2 create hyperlinks in web pages

CC and BCC in Email (Code)

Instructor Notes

```
<!DOCTYPE html>
<html>
<body>
<!--add the CC and BCC's-->
<a
href="mailto:edude@capgemini.com?Subject=Hello&Cc=jane@capgemini.com&
bcc=joe@capgemini.com"> Mail us </a>
</body>
</html>
```

Capgemini Public



Links to Specific Part of a Page

Instructor Notes

- id attribute, used in the anchor tag identifies a section of a page.
- Set HREF to the section name. Precede the name with a # symbol

```
<a id="address1"></a>
```

Capgemini Ltd, Rajiv Gandhi Infotech Park, Hinjewadi Phase III MIDC
Pune 411 057. INDIA.

... Contents of the page...

I am working in Capgemini

Capgemini Public



For documents located on other computers, attach the section name, preceded by the # symbol, at the end of the document URL.

E.g. I am working in

```
<a href="http://www.state.edu/add.htm#address1">Capgemini</a>
```

Create Link to A Specific Part of A Page

ID attribute, used in the anchor tag, identifies a section of a page. Subsequently, users can access this section via a link within the document or other documents.

To include a link to the named section elsewhere in your page, create a link and set **href** to the name of the section as defined by the ID attribute. Precede the name with a # symbol to differentiate it from the name of another document.

For example:

```
<a id="section1"></a>
```

```
< -- Text of section 1-- >
```

You find the relevant information in

```
<a href="#section1">section 1</a>.
```

To include a link to the named section from another local document, create a local link and include the name of the section, preceded by the # symbol, in the location of the link defined by the **href** attribute.

Demo

- Linkall.html



Capgemini Public



```
<!DOCTYPE html>
<html>
<head>
<title>Using links</title>
</head>
<body >
<a href="6image1.html" target="_blank" > link to a local page</a>
<p>
<a href="http://eclassroom.igate.com" > link a site</a>
<p>
<p>
<a href="mailto:eDude@igate.com?subject=More Info"> mail</a>
your comments to me
</body>
</html>
```

Providing Target for a Hyperlink

Instructor Notes

- **Target attribute of the <a> tag specifies where to load the linked document.**
`<a href="link2.htm" target="frame2">Document 2</a>`
- **It takes following values:**
 - name of the existing window/target
 - `_blank`
 - `_self`
 - `_parent`
 - `_top`
- **By default it loads the linked document in the current window.**

_blank (Code)

- “_blank”:
 - opens a new document in a new window.
- This code produces this:

```
<a href="newwindow.html" target="_blank">a new window</a>
```

Using target="_blank"

This causes the link to open in a totally new browser window, with the page with the referring link still open behind it. Unlike JavaScript pop ups, the developer has no control over the size of the resulting window. It depends on what the browser did the last time before it was closed.

Instructor Notes

4.2 create hyperlinks in web pages

_self (Code)

- “_self”:
 - Puts new document in the same window and frame as the current document.
 - Works the same as if you had not used TARGET at all.
- This code produces this go to:

```
<a href="selftarget.html" target="_self">next</a> page
```

Capgemini Public



Using target="_self"

This loads the page within the same frame as the link tag.

Instructor Notes

4.2 create hyperlinks in web pages

_top (Code)

- “_top”:
 - loads linked document in the topmost frame. This means that the new page fills the entire window.

```
<a href="selftarget.html" target="_top">top</a>
```

Capgemini Public



Using target="_top"

This, within a link tag causes the new page to load in the full body of the window. It is useful if you want to break out of the frameset you have created in order to have a frameless page.

Instructor Notes

4.2 create hyperlinks in web pages

_parent (Code)

- “_parent”:
 - Used when a frameset file is nested inside another one
 - A link in one of the inner frameset documents which uses “_parent” loads the new document where the inner frameset file had been

```
<a href="bigframe.html" target="_parent">bigframe</a>
```

Capgemini Public



Using target="_parent"

This is similar to target="_top" but refers to the immediate parent of a frame. In more advanced frame usage there may be several nested frames. This allows more control over which frames are specified. (It is actually something developers rarely need to use).

Instructor Notes

4.2 create hyperlinks in web pages

Demo

- `Linktarget.html`



Capgemini Public



4-3: Add hyperlinks to list items and table contents

Hyperlinks in Lists Items (Code)

- Add links to list items and table contents:

```
<ul>
  <li><a href=home.html>mumbai</a></li>
  <li><a href=home.html>pune</a></li>
  <li><a href=home.html>nasik</a></li>
</ul>
```

Capgemini Public



4-3: Add hyperlinks to list items and table contents

Hyperlinks in Table Elements (Code)

Instructor Notes

- Add links to table contents:

```
<table border=1>
<tr><th>team<th>points<th>grade</tr>
<tr><td>
<a href=home.html>mumbai</a></td><td>90</td><td>a</td>
</tr>
<tr><td>
<a href=home.html>pune</a></td><td>86</td><td>b</td>
</tr>
<tr><td>
<a href=home.html>nasik</a></td><td>80</td><td>c</td>
</tr>
</table>
```

Capgemini Public



4.3: Add hyperlinks to list items and table contents

Demo

- Linkintable.html



Capgemini Public



```
<!DOCTYPE html>
<html>
<head>
<title>Links in lists and table Example</title>
</head>
<body>
<ul>
<li><a href=home.html>mumbai</a>
<li><a href=home.html>pune</a>
<li><a href=home.html>nasik</a>
</ul>
<table border=1>
<tr><th>team<th>points<th>grade
<tr><td><a href=home.html>mumbai</a><td>90<td>a
<tr><td><a href=home.html>pune</a><td>86<td>b
<tr><td><a href=home.html>nasik</a><td>80<td>c
</table>
</body>
</html>
```


Instructor Notes

LAB

- Lab 4



Capgemini Public



Summary

Instructor Notes

- In this lesson, you have learnt about
 - How hyperlink works in web pages
 - Creation of hyperlinks in web pages
 - Including hyperlinks to list items and table contents



Capgemini Public



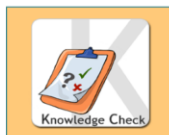
Instructor Notes

Review Question

Answers

- 1: Page Section
- 2: False
- 3.target

- Question 1: The id attribute along with # used in the anchor tag identifies:
 - Page section.
 - Path to load the linked document.
 - Address to send the e-mail to.
- Question 2: URL tag mailto, identifies the address from where e-mail is to be received.
 - True/ False
- Question 3: _____ attribute of the <a> tag specifies where to load the linked document.



Capgemini Public



Review Question: Match the Following

Instructor Notes

Answers

- 1 – e
- 2 – d
- 3 – f
- 4 – a
- 5 – b
- 6 – c

| | |
|---------------------|---|
| 1. “_blank” | a. Identifies address to send the e-mail to. |
| 2. “_self” | b. Only filename is needs to be specified. |
| 3. “_top” | c. Gives entire address of the page. |
| 4. mailto | d. Puts the new document in the same window and frame as the current document |
| 5. Relative Linking | e. Opens the new document in a new window. |
| 6. Absolute Linking | f. Loads linked document in the topmost frame. |

Capgemini Public



Instructor Notes

Web Basics – HTML5

Lesson 5. Image Handling

Capgemini Public



Instructor Note:

Lesson Objectives

➤ **After completing this module, you will be able to:**

- Understand the role of images in web pages
- Learn to add images to web pages
- Learn to use images as hyperlinks



Capgemini Public



Instructor Note

5.1: Role of images in web pages

Roles of images in web pages

- **Web pages are comprised of text and images.**
- **Some of the roles of images in web pages are**
 - To make web page content more interesting to be read
 - To increase clarity on web page content by transmitting right message
 - Provoking reactions
 - To ensure that the user response is anticipated
- **Some of the widely used image formats**
 - GIF : Graphical Interchangeable Format
 - JPEG : Joint Photographic Expert Group
 - PNG : Portable Network Graphics

Capgemini Public



GIF images are quickly downloadable, because its size is less. It is used for animated images, thumbnails, B/W image . It supports only 256 colors

JPEG images uses [lossy](#) compression method, meaning that some original image information is lost and cannot be restored, possibly affecting image quality.

PNG images are lossless, portable, well-compressed storage of raster images. It supports image transparency

Instructor Note

5.2 Adding images in webpages

Adding images in webpages

- Images are added into a HTML document using tag.
- tag has 2 required attributes such as src and alt
- Syntax:

```

```

- The src attribute defines the url (web address) of the image
- Alt attribute defines alternative text to display for non-graphical browsers. The alt attribute is required

- Example:

```

```

Capgemini Public



GIF images are quickly downloadable, because its size is less. It is used for animated images, thumbnails, B/W image . It supports only 256 colors

JPEG images uses [lossy](#) compression method, meaning that some original image information is lost and cannot be restored, possibly affecting image quality.

PNG images are lossless, portable, well-compressed storage of raster images. It supports image transparency

Instructor Note:

5.2: Adding Images in webpages

Attributes of an Inline Image

- **Other attributes of tag are:**
 - width: Sets the width of the image.
 - height: Sets the height of the image.

Capgemini Public



After you have the image, save it in GIF format. Then, use the tag to include the image in your document. The tag is empty, which means that it contains attributes only and it has no closing tag.

```

```

Following list shows attributes and options you can include inside the tag:

alt: Alternative text to display for non-graphical browsers

align: Aligns graphic with surrounding text. Values include left, right, and center

vwidht: Spacing above and below an image and surrounding text.

hwidth: Spacing between an image and surrounding text.

Border: Sets a border around a graphic. It takes a numerical value to determine border thickness. A higher number indicates a thicker border.

width: Sets the width of the image.

height: Sets the height of the image.

Instructor Notes

5.2: Adding Images in webpages

Text & Image Alignment (Code)

➤ **Style Attribute:**

- Can be used in tag to define the alignment of text and image.

| Style Attribute Value | Description |
|-----------------------|---|
| vertical-align:bottom | The bottom of the element is aligned with the lowest element on the line |
| vertical-align:middle | The element is placed in the middle of the parent element |
| vertical-align:top | The top of the element is aligned with the top of the tallest element on the line |
| float:right | The element floats to the left |
| float:left | The element floats to the right |
| float:none | The element floats to the none |

- **<br clear=LEFT/RIGHT>** can be used to stop text flowing around an image

Capgemini Public



Insert the <br clear=LEFT/RIGHT> command at the spot in the descriptive text where text is to stop flowing around the image.

Demo

- ImageDemo.html
- ImageAlign.html

The Example of image alignment



HTML allows for simple aligning of text and images by adding the style attributes to the tag. HTML 5 expanded this attribute by specifying float:left and float:right options. When you use these options, text flows down the right or left side of an image aligned to them. The text flow continues until either the text passes the image, the text ends, or the browser encounters a tag.

HTML5 allows vertical alignment of text and images by adding the style attribute with values as vertical-align:bottom, vertical-align:top,... When you use these options, text is vertically aligned top/bottom/middle. The text flow continues until either the text passes the image, the text ends, or the browser encounters a br tag with clear attribute.



Instructor Note:

5.3: Image as hyperlinks

Use of Image as a Hyperlink (Code)

- Images used as hyperlinks:

```
<a href="6image1.html"></a>
```

- Images contained within a table:

```
<table>
<tr><th>Product</th><th>Cost</th><th>Image</th></tr>
<tr> <td>Pencil</td> <td>$8</td>
      <td></td> </tr>
<tr> <td>Brush</td> <td>$15</td>
      <td></td></tr></table>
```

Capgemini Public



Use Image as Link

To create an image link, place the tag with the image URL between tags.

```
<h2>MY FIRST IMAGE LINK</h2>
<a href="BALLOON.HTM"></a>Click on it to have
additional information on it.
```

Instructor Notes

5.3: Image as hyperlinks
Demo

- ImageLink.html



Instructor Notes

Lab

- Lab 5



Capgemini Public



Instructor Notes

Lesson Summary

- In this lesson, you have learnt about:
 - Role of images in web pages
 - How to add images in web pages
 - How to use images as hyperlinks



Capgemini Public

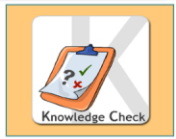


Instructor Notes

Answers
1:

Review - Questions

- Question 1: Client-side image maps are usually faster than server-side image maps.
 - True/ False
- Question 2: _____ contain the linking information on the server.



Instructor Note:

Web Basics – HTML5

Lesson 6. HTML Forms for User Input

Capgemini Public



Instructor Note

Lesson Objectives

- After completing this module you will be able to:
 - Understand the role of forms in web pages
 - Understand various HTML elements used in forms.
 - Single line text field
 - Text area
 - Check box
 - Radio buttons
 - Password fields
 - Pull-down menus
 - File selector dialog box



Capgemini Public

 **Capgemini**
CONSULTING TECHNOLOGY OUTSOURCING

Instructor Note

6.1 Role of forms in web pages

HTML Forms for User Input

- HTML forms are used to accept user inputs and then submit data for processing.
- A *form* is an area that contains form elements.
- Types of elements which can be included in a form are
 - Label
 - Single line text field
 - Password field
 - Text area
 - Drop down menu
 - Radio button
 - Checkbox
 - File selector box, etc..

Capgemini Public



Forms:

HTML forms are used to accept of user input.

A *form* is an area that contains form elements.

Form elements are elements that allow users to enter information

(text fields, text area fields, drop-down menus, radio buttons, checkboxes, etc.) in a form.

Define a form with the <form> tag.</form>

Instructor Note:

6.1 Role of forms in web pages

HTML Forms for User Input

- User input forms are created using `<form>` tag.
- Syntax:

```
<form method="get/post" action="URL" enctype="Encryption Type">
    Field definitions
</form>
```

- **action:** the URL of the script
- **method:** the HTTP request method to use, sometimes GET, but usually POST
- **enctype:** Specifies how the data is to be encoded.

Capgemini Public



Basic syntax for the `<form>` tag is:

```
<form method="Get or Post" action="URL" enctype="type">
    Field definitions
</form>
```

`<form>` tag tells a browser that there is a *fill-in-the-blank* form in this HTML document.

Method:

method attribute states the method to use when you send the form to the server. Two acceptable methods are GET and POST.

GET sends information entered in the form to the server at the end of the URL.

POST sends information entered in the form to the server as a data body/document.

action attribute:

Gives the address of the script that processes the form.

Defines the name of the file to send the content to. File defined here typically does something with the received input.

enctype attribute:

Specifies how the data is to be encoded.

Applies only if you use the POST method. There is only one possible value, the default value "application/w-www/form-urlencoded".

HTML Forms for User Input

Instructor Note

- Some more attributes which can be used in <form> tag are

| Attribute Name | Attribute Value | Description |
|----------------|------------------------------|--|
| name | Form name as a string | Mentions the name of a form. |
| autocomplete | On, off | Specifies whether a form should have autocomplete on or off |
| target | _blank, _self, _parent, _top | Specifies where to display the response that is received after submitting the form |
| novalidate | novalidate | Specifies that the form should not be validated during form submission. |

Instructor Note

6.2 Various HTML elements used in forms

HTML Form Elements

- `<input>` element is the most used form tag.
- An `<input>` tag includes the following attributes
 - name: Name of the field which is required to send data(Key/Value pair) during form submission
 - id: A unique identified of the field
 - value: Sets a default value of the field
 - maxlength: Specifies the maximum number of characters allowed in an `<input>` element
 - readonly: Specifies that an input field is read-only
 - size: Specifies the width, in characters, of an `<input>` element
 - Disabled: specified that an input element should be displayed.
 - type attribute of `<input>` tag specifies the field type

Capgemini Public



Input:

The most used form tag is the `<input>` tag. Input type is specified with the type attribute. Type attributes are:

- Text
- Password
- Hidden
- Radio
- Checkbox
- File
- Button
- Submit/reset

Instructor Note:

6.2 Various HTML elements used in forms

Text-related Elements

- Text related elements can be created as shown below:

| Code | Element |
|--|----------------------|
| <code><input type="text"></code> | Single line text box |
| <code><input type="password"></code> | Password field |
| <code><input type="hidden"></code> | Hidden field |

- Multiple line text input control**
 - If input exceeds more than one line, then create Multi-line input control using HTML `<textarea>` tag
 - Syntax:

```
<textarea rows=" " cols=" " name=" ">
```

 - Rows : Number of rows of text area box
 - Cols: Number of columns of text area box
 - Name: name of the element

Capgemini Public



Instructor Note:

6.2 Various HTML elements used in forms

Checkbox Element

- If more than one option is required to be selected from multiple options, then create checkbox as shown below:
 - `<input type="checkbox">`
 - Use checked attribute for selecting any checkbox to be selected by default
- Example:

```
<input type="checkbox" name="hobbies" value="Reading Books"> Reading Books  
<input type="checkbox" name="hobbies" value="Net Surfing"> Net Surfing  
<input type="checkbox" name="hobbies" value="Singing" checked> Singing
```

Select your Hobbies: ☐ Reading Books ☐ Net Surfing ☒ Singing

Capgemini Public



Instructor Note:

6.2 Various HTML elements used in forms

Radio Button

- If only one option is required to be selected from multiple options, then create radio button as shown below:
 - `<input type="radio">`
 - Use checked attribute for making a radio button to be selected by default
- Example:

```
<input type="radio" name="sector" value="Public"> Public  
<input type="radio" name="sector" value="Private"> Private
```

Select your sector in which you are working: ☐ Public ☐ Private

Capgemini Public



Instructor Note

6.2 Various HTML elements used in forms

Drop down list

- Drop down list allow the user to select one or more values from a pre-determined options
- Tags for creating drop down list with options are:
 - <select> : Creates drop down list
 - <option>: Defines an option in a select list.

| Tag name | Attribute | Description |
|----------|-----------|---|
| <select> | Name | Defines a name for the drop down list |
| | Size | Defines the number of visible options in a drop down list |
| | Multiple | Allow to select multiple options at once |
| | Disabled | Disable drop down list |
| <option> | value | Specifies the value to be sent to a server |
| | Selected | Makes option to be selected by default |

Capgemini Public



Drop down list - Example

Instructor Note:

```
<!DOCTYPE html>
<html>
<body>Select a country:
  <select name="country">
    <option value="Germany">Germany</option>
    <option value="India" selected>India</option>
    <option value="China">China</option>
    <option value="Japan" >Japan</option>
  </select>
</body>
</html>
```

Select a country: ▼

Capgemini Public

Instructor Note

6.2 Various HTML elements used in forms

File Upload

- File upload will allow the user to upload a file from the desktop to an application in browser.
- The below code is used to define a file-select field and a "Browse..." button (for file uploads):
 - `<input type="file"/>`

| Attribute | Description |
|-----------|---|
| Name | Defines a name for the file upload dialog box |
| Disabled | Disable element |
| Accept | Specify MIME type to describe the file type which accepts by a server |

- In `<form>` tag, use `enctype="multipart/form-data"` if file need to be uploaded using file selector form element.

Capgemini Public



File Upload- Example

Instructor Note:

```
<!DOCTYPE html>
<html>
<body>
  <form method="post" action="success.html"
    enctype="multipart/formdata">
    Select a photo to upload:
    <input type="file" name="photo"/>
  </form>
</body>
</html>
```

Select a photo to upload:

Capgemini Public



Instructor Note

6.2 Various HTML elements used in forms

Button

- Different types of button which is possible to be created in HTML5 are as shown below:

| Field type | Element | |
|-----------------------|---------------|--|
| <input type="button"> | Button | A clickable button, that activates a JavaScript when it is clicked |
| <input type="submit"> | Submit button | Defines a button for submitting a form |
| <input type="reset"> | Reset button | Define a reset button (resets all form values to default values) |

Capgemini Public



Instructor Notes:**Single Line Text Field**

`<input>` tag: Add an `<input>` tag to accept some input from the reader. This is a standalone tag. This is the default input type .Syntax is as follows:

```
<input type="text" name="UserID" size=20
value="Guest" maxlength=unlimited [disabled] [readonly]>
```

Give the user the option to submit this form or reset it and enter fresh values. To do this, use the `<input>` tag and set **type** to either "SUBMIT" or "RESET".

```
<input type="submit" name="submit_button" value="Submit">
<input type="reset" name="reset_button" value="Reset">
```

type attribute tells the browser the type of button used. **name** is a variable that you can access later when referring to this information. **value** is button caption.

Text Fields: These are used to type letters, numbers, etc. in a form.
Example:

```
<form>
First name: <input type="text" name="firstname">
<br>
Last name:<input type="text" name="lastname">
</form>
```

Text Area:

The `<textarea>` tag defines a multi-line text input control.

A text area can hold an unlimited number of characters. Text renders in a fixed-width font (usually Courier). You can specify text area size with *cols* and *rows* attributes. An even better way to do it is through CSS *height* and *width* properties.

```
<textarea rows="4" cols="20">
<textarea name="name" rows="10" cols="50" [disabled][readonly]>
Default Text
</textarea>
```

```
<textarea name="address" rows=5 cols=10>
Please write your address</textarea>
```

- [CheckboxExample.html](#)
- [FormExamples.html](#)
- [PasswordField](#)
- [RadioButtonExample](#)
- [RadioExample](#)
- [TextFieldExample.html](#)



CONSULTING. TECHNOLOGY. OUTSOURCING.

Page 0-140

Instructor Notes:

```

<p>
<b>Select the stream you belong to :</b> <input
type="radio" value="science" name="s-grp"> Science <input
type="radio" value="arts" name="s-grp"> Arts <input
type="radio" value="commerce" name="s-grp"> Commerce <input
type="radio" value="oth2" name="s-grp">Engineering
</p>

<b>Which training program would you like to attend ?</b> <select
multiple size="3" name="pref">
<option value="ih" selected>Internet-HTML
<option value="js">Javascript
<option value="vbs">VBscript
<option value="as">ASP
<option value="xm">XML
<option value="jv">JAVA
<option value="jsp">jsp</option>
</select> <br>

<br> <b>Select the location of your resume</b> <input
type="file" size="15" name="fnm"> <br> <input
type="hidden" value="patni" name="coname"> <br> <input
type="button" value="exit" name="but"> <input type="submit"
value="save"> <input type="reset" value="reset">

</form>
</body>
</html>

```

Instructor Note:

Demo

- FormSubmission.html



Capgemini Public



```
<!DOCTYPE html>
<html>
<head>
<title>Form Submission</title>
</head>
<body>
<form method="post" action="store.html">
<p>Enter UserID:<input id=text1 name=text1></p>
<p>Enter Pasword:
<input id=password1 type=password name=password1></p>
<p>
<input id=submit1 type=submit value=Submit name=submit1> &nbsp;
<input id=reset1 type=reset value=Reset name=reset1></p>
</form>
</body>
</html>
```

Instructor Note:

Lab

- Lab 7



Capgemini Public



Instructor Note:

Summary

- **After completing this module you now:**
 - Understand the role of forms in web pages
 - Understand various HTML elements used in forms like:
 - Single line text field
 - Text area
 - Check box
 - Radio buttons
 - Password fields
 - Pull-down menus
 - File selector dialog box



Capgemini Public



Instructor Note:

Review Question

- Question 1: Radio Buttons are used when you want the user to select:
 - Option 1: one of a limited number of choices.
 - Option 2: one or more options of a limited number of choices.
 - Option 3: many of unlimited number of choices.
- Question 2: METHOD attribute states the method to use when you send the form to the server.
 - True/ False
- Question 3: The _____ attribute of form specifies how the data is to be encoded.



Capgemini Public



Instructor Note:

Review Question: Match the Following

Answers:

1. F
2. E
3. D
4. C
5. B
6. a

1. Check Boxes

2. Radio Buttons

3. Password fields

4. Hidden Input Type

5. <select> tag in
Drop Down List6. <option> tag in
Drop Down List

a. Populate data in the list.

b. Create a select list.

c. Include form data without
rendering it to the user.d. Entered text is shown as
asterisks.e. Only one option of a limited
number of choices.f. One or more options of a limited
number of choices

Capgemini Public



Add the notes here.

Instructor Notes:



Web Basics – HTML5

Lesson 7. New Form Elements

Capgemini Public

Instructor Notes:

Lesson Objectives

- **In this lesson you will learn about:**
 - Understand the new HTML form elements such as
 - date
 - number
 - range
 - email
 - search
 - datalist
 - Understand audio, video, article tags

Capgemini Public

Introduction to HTML5 Enhanced Form

- HTML5 brings to the table several new input types, a total of 16
- HTML5 introduces these data types via the `<input type="_NEW_TYPE_HERE_"/>` format
- One of the key design decisions in HTML5 is backward compatibility
- It provides automatic validity of the fields as per the form's constraints
- The browser inspects the input type and if it finds that it is not supported, it falls back to the default type

- a) You get automatic validity of the fields as per the format. This means that the form is not going to get submitted if the value entered is not as per the default validation of that type
- b) The browser inspects the input type and if it finds that it is of a specific type, then it does something quite clear to aid the input of that data. For e.g. On the Smart Phones, which do not have a physical keyboard but instead a virtual keyboard, the keyboard that will be shown up will only contain keys that will aid the user in filling out the data.

Instructor Notes:

7.1 : New HTML form elements

Placeholder

- **Place Holder** - A placeholder is a textbox that hold a text there is no value and not focused
- **Syntax is –**

```
<input id="first_name" placeholder="This is a placeholder">
```
- **This is how place holder looks like on supporting browser**

- **Once the textbox gets focus, the text goes off and you can enter your text**


Firefox 3.7 Safari 4.0 Chrome 4.0 Opera

Capgemini Public

Place Holder:

The first improvement HTML5 brings to web forms is the ability to set placeholder text in an input field. Placeholder text is displayed inside the input field as long as the field is empty and not focused. As soon as you click on (or tab to) the input field, the placeholder text disappears.

Here's how you can include placeholder text in your own web forms:

```
<form>
```

```
<input name="name" placeholder="Enter your name">
```

```
<input type="submit" value="Search">
```

```
</form>
```

Browsers that don't support placeholder attribute will simply



Instructor ignore it. But if you want to make it work in other browsers, you can use some **JavaScript** to create the same behavior. There is an excellent **jQuery plugin** called **HTML5 Placeholder** Plugin that will go through all input fields with placeholders attached to them and make them work in all browsers.

Instructor Notes:

7.1 : New HTML form elements

Auto Focus

- **Auto Focus** - Autofocus is a Boolean attribute of form elements that set focus on it when a page is loaded
- **Syntax is –**

```
<input id="Text2" type="text" autofocus/>
```

Firefox 4 Safari 4.0 Chrome 3.0

Capgemini Public

Auto Focus

This attribute when applied to any form element, will result in the field receiving focus.

For e.g. consider the example shown below:

<form>

<label for="firstname">First Name</label>

<input type="text" id="firstname" name="firstname" autofocus>

<label for="lastname">Last Name</label>

<input type="text" id="lastname" name="lastname">

<input type="submit" label="Go">

</form>

We have added the attribute autofocus to the firstname input field. When the form

loads, you will find that the focus is already set on that field, thereby making it easier



Instructor for the user to start filling the form.

Instructor Notes:

7.1 :New HTML form elements

Required

- Required - A field with “required” attribute must be before submission of a form
- Syntax is –

```
<input name="name" type="text" required />
```

- The picture below shows us how Firefox and Opera p if a "Required field" is left blank upon submission

What is your favorite movie : Submit

Please fill out this field.

What is your favorite movie : Submit

This is a required field

Firefox 4

Capgemini Public

Required:

If the **required** attribute is present, then the field must contain a value when the form is submitted. This informs the (HTML5-aware) web browser that the field is to be considered mandatory. Different browsers may mark the input box in some way (Firefox 4 Beta adds a red box-shadow by default), display a warning (Opera) or even prevent the form from being submitted if this field has no value. Hopefully these behaviours will converge in future releases.

Here's an example of an input field for a required email address that ensures that the field has a value and that the value is a valid email address.

Example:



Instructor `<input type="email" id="email_addr" name="email_addr" required />`

Instructor Notes:

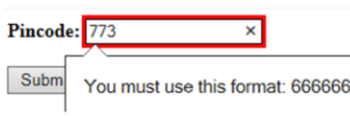
7.1 : New HTML form elements

Pattern

- **Pattern - A value filled in the field must be checked against the regular expression specified in pattern attribute.**
- **Syntax is –**

Pincode: `<input type="text" name="pin_code" pattern="[0-9]" />`

- **The picture below shows us how browser prompt us “field with Pattern attribute” is filled with invalid value.**



Pincode: 773 x

Submit You must use this format: 666666

Capgemini Public

Required:

If the **required** attribute is present, then the field must contain a value when the form is submitted. This informs the (HTML5-aware) web browser that the field is to be considered mandatory. Different browsers may mark the input box in some way (Firefox 4 Beta adds a red box-shadow by default), display a warning (Opera) or even prevent the form from being submitted if this field has no value. Hopefully these behaviours will converge in future releases.

Here's an example of an input field for a required email address that ensures that the field has a value and that the value is a valid email address.

Example:



Instructor `<input type="email" id="email_addr" name="email_addr" required />`

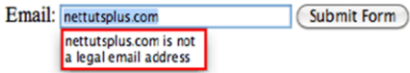
Instructor Notes:

7.1 : New HTML form elements

Email

- **Email** - This field is used to check whether the string is a valid email id or not.
- **Syntax is –**

```
<input id="email" name="email" type="email" />
```
- **Browsers that don't support this field will treat this as a text field.**
- **This is how it looks like on form**



Capgemini Public

Email:

The email type is used for input fields that should contain an e-mail address. The value of the email field is automatically validated when the form is submitted.

Example

E-mail: `<input type="email" name="user_email" />`

Safari on the iPhone recognizes the email input type, and changes the on-screen keyboard to match it (adds @ and .com options).

Instructor Notes:

7.1 : New HTML form elements

Data List

- Data List - Datalist is seems like type-ahead auto suggest in Google search box
- Syntax is –

```
<input id="country_name" name="country" type="text" list="country" />
<datalist id="country">
  <option value="Australia">
  <option value="Austria">
  <option value="Algeria">
  <option value="Andorra">
  <option value="Angola">
</datalist>
```

**Data List:**

The datalist element specifies a list of options for an input field. The list is created

with option elements inside the datalist. To bind a datalist to an input field, let the list

attribute of the input field refer to the id of the datalist:

Example

```
Webpage: <input type="url" list="url_list" name="link" />
<datalist id="url_list">
  <option label="W3Schools"
    value="http://www.w3schools.com"/>
  <option label="Google" value="http://www.google.com"/>
  <option label="Microsoft"
    value="http://www.microsoft.com"/>
</datalist>
```

Instructor Notes:

7.1 : New HTML form elements

Search

➤ **Search** - In HTML5, we can define a textbox as search textbox

- Supported by Chrome 8 & Safari 5
- Syntax is

```
<input id="mysearch" type="search" />
```



Capgemini Public

Search:

Setting the type attribute of an **<input>** tag to **search** is mainly of cosmetic value. It

doesn't automatically create a search field. As with all form elements, you still need

to code the server-side logic yourself or use a third-party script. Most browsers

simply display a search field as a normal text field. However, Safari and Chrome on

Mac OS X automatically add rounded corners to match the operating system's

standard look.

Apple.com uses **<input type="search">** for their site-search box, to help give their

Site a "Mac-like" feel. But there's nothing Mac-specific about it. It's just markup, so

each browser on each platform can choose to render it according to platform-specific

conventions. As with all the other new input types, browsers that don't

recognize **type="search"** will treat it like **type="text"**.



Instructor

Example:

```
<form>  
  <input name="q" type="search">  
  <input type="submit" value="Find">  
</form>
```

Instructor Notes:

7.1 : New HTML form elements

Number

- Up and down button provided to increase and decrease
- Min and max parameters provided to limit the values
- Browser will treat it as simple textfield if it doesn't support
- Syntax is

```
<input id="movie" type="number" value="0"/>
```

- Limiting the values for this field...

```
<input id="userLIC" type="number" min="5" max="30"/>
```

Number:

Asking for a number is trickier than asking for an email address or web address.

First of all, numbers are more complicated than you might think. You don't often

ask for "just a number." It's more likely that you'll ask for a number in a

particular range. You may only want certain kinds of numbers within that range-

maybe whole numbers but not fractions or decimals.

Example:

```
<input type="number" min="0" max="10" step="2" value="6">
```

Let's take that one attribute at a time.



Instructor

1. type="number" means that this is a number field.
2. min="0" specifies the minimum acceptable value for this field.
3. max="10" is the maximum acceptable value.
4. step="2", combined with the min value, defines the acceptable numbers in the range: 0, 2, 4, and so on, up to the maxvalue.
5. value="6" is the default value.

Instructor Notes:

That's the markup side of a number field. Keep in mind that the `min` and `max` attributes are optional. If you have a minimum but no maximum, then you can omit the `max` attribute. The default step value is 1. If you omit the `step` attribute unless you need a different step value, then the `value` attribute can be the empty string, `''`, altogether.

Visualizing Number field :

On the iPhone, where input is difficult to begin with, the browser once again optimizes the virtual keyboard for numeric input.

In the desktop version of Opera, the same `type="number"` control, with little up and down arrows that allow you to increment or decrement the value.

Opera respects the `min`, `max`, and `step` attributes, so you can't enter an unacceptable numeric value. If you bump up the value to the maximum, the up arrow in the spinbox is greyed out.

Instructor Notes:

7.1 : New HTML form elements

Range

- Also known as slider
- Before HTML5, programmers used to write lines of c
- Easy to implement
- Look and feel is different for different browser's
- Browser will treat it as simple textfield if it doesn't s
- Syntax

```
<input id="test" type="range"/>
```

The image shows three different browser interfaces for the HTML5 range slider. Safari 5 shows a simple horizontal line with a vertical slider knob. Chrome 8 shows a more detailed slider with a vertical knob and a horizontal track. Opera shows a similar slider with a vertical knob and a horizontal track. The browsers are labeled below their respective UIs: Safari 5, Chrome 8, and Opera. The text 'Capgemini Public' is visible at the bottom right of the slide.

Range:

Spinboxes are not the only way to represent numeric input.

You've probably also seen "slider" controls.

You can now have slider controls in your web forms, too.

Example:

```
<input type="range" min="0" max="10" step="2" value="6">
```

All the available attributes are the same as `type="number"` — `min`, `max`, `step`, `value` — and they mean the same thing. The only difference is the

user interface. Instead of a field for typing, browsers are expected to

render `type="range"` as a slider control. Safari, Chrome, and Opera all do this.

(Sadly, the iPhone renders it as a simple text box. It doesn't



Instructor even optimize its on-screen keyboard for numeric input.) All other browsers simply treat the field as type="text".

Instructor Notes:

7.1 : +New HTML form elements

Date

➤ **Date**

- Important and mostly used element
- Simple to implement
- Before HTML5, programmers used to write lines of java
- Input type for date:- date, week, month, time, datetime local (local time)

➤ **Syntax is**

```
<input id="meeting" type="date" value=""/>
```



Capgemini Public

Date:

HTML 4 did not include a date picker control. JavaScript frameworks have picked up

the slack (Dojo, jQuery UI, YUI, Closure Library), but of course each of these

solutions requires “buying into” the framework on which the date picker is built.

HTML5 finally defines a way to include a native date picker control without having to

script it yourself. In fact, it defines six input types: date, month, week, time, date + time, and date + time - timezone.

So far, support is... sparse.

Instructor Notes:

7.2 : Understand audio, video, article tags

Audio

Audio - HTML5 is likely to put an end to audio plug-in such as Windows Media player, Microsoft Silverlight , Apple QuickTime, and the famous Adobe Flash

- MIME type's - *audio/mpeg*, is optional but its always better
- Only .mp3, .wav, and .ogg (vorbis) formats are supported
- If quick time player is not available, then safari won't support
- Other properties like auto play, loop, preload area also supported
- Syntax is –

```
<audio controls>
<source src="vincent.mp3" type="audio/mpeg"/>
<source src="vincent.ogg" type="audio/ogg"/>
</audio>
```

Audio & Video:

Capgemini Public

HTML5 is fairly intelligent about picking the right default for presenting the most

optimum audio or video. Couple that with the absolute minimum coding that's

needed to handle multimedia in HTML5 and you have a pretty good situation for

developers.

The bad news is that because the people diligently working on the HTML5

specification tried to compromise between open formats and de facto standard

formats and so on, support for native codecs in HTML5 is slightly lacking: there

isn't any. It's up to the browser to support formats, and up to the developer to

supply them. What's emerged from that are a few relatively new standards.

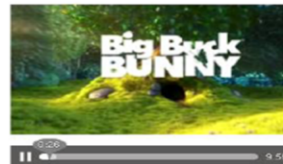
Instructor Notes:

7.2 : Understand audio, video, article tags

Video

- **Video – HTML5 video tag is exactly similar to audio b attributes**
- **Attributes**
 - Width : Width of video area
 - Height : Height of video area
 - Poster : Still Image file projected on screen before video
- **Syntax is –**

```
<video src="http://.....ogv" controls width="300" he
```



Examples:

Capgemini Public

Audio

```
<audio controls preload="metadata">
```

```
<source src="sample.mp3">
```

```
<source src="sample.ogg"> </audio>
```

Video

```
<video controls width="640" height="480" src="sample.mp4"
poster="sample.jpg">
```

```
</video>
```

Both the <video> and <audio> can take another important parameter, shown

above, called preload (which was formerly implemented as autobuffer with

slightly different syntax). The values for preload can be auto



Instructor

(download the media file to the browser in advance), none (do not preload the media), or metadata (just download enough metadata to discover the duration and other information of the media file). So with a little extra care in coding, you can make the user's experience much better.

Instructor Notes:

7.2 : Understand audio, video, article tags

Article

➤ Article

- Represents independent component on page
- Separates itself from other elements on page
- Can be used most effectively in forum posts, blogs, magazines

➤ Syntax is

```
<article>
<header>
<h1>The Very First Rule of Life</h1>
<p>Published : <time pubdate="pubdate">2009-
</header>
<p>If there's a microphone anywhere near you,
and sending whatever you're saying to the world
<p>...</p>
<footer>
<a href="?comments=1">Show comments...</a>
</footer>
</article>
```

Article:

Capgemini Public

The article element represents a component of a page that consists of a self-contained composition in a document, page, application, or site and that is intended to be independently distributable or reusable, e.g. in syndication. This could be a forum post, a magazine or newspaper article, a Web log entry, a user-submitted comment, an interactive widget or gadget, or any other independent item of content.

Instructor Notes:

Demo

- Demonstration on the use of new form elements



		Capgemini Public
--	--	------------------

Instructor Notes:

Lab

- Lab 8

Capgemini Public

Instructor Notes:

Summary

- **In this lesson, you have learnt about:**

- HTML5 introduces new and enhanced form elements those caters to every need of modern Web Designing
- It also introduces a new JavaScript library, Modernizr, that helps you detect HTML5 feature capabilities of your websites

Capgemini Public

Instructor Notes:

Answers

1. Placeholder
2. Modernizr

Review Question

- Question 1: A _____ is a textbox that hold a text lighter shade when there is no value and not focused
- Question 2: A _____ is a JavaScript library that you add HTML5 capabilities to your websites

Capgemini Public